

VolTree Architecture

1 INPUTS



Market data

OHLCV → Parkinson variance



Earnings dates

External calendar

2 FEATURES

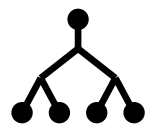
Endogenous (8)

HAR lags + momentum

Earnings (4)

Distance to release

3 XGBOOST FORECAST



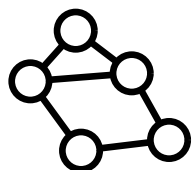
Gamma-loss XGBoost

300 trees · 31 leaves · reg:gamma ≈ QLIKE

Per-stock forecast

$\hat{\sigma}_{i,t+h}^2 \cdot h \in \{1, 5, 10, 22\}$

4 GRAPH REFINEMENT



Leaf-cooccurrence graph smoothing

kNN on shared leaves ($k = 10$)

$$\hat{y}^{sm}_i = (1 - \alpha) \hat{y}_i + \alpha \bar{y}_{knn}^{(i)}$$

α fit on validation